



南开大学
Nankai University

南 开 大 学

网 络 空 间 安 全 学 院

区块链 Lab2 实验报告

涉及四方的多签名交易

姓名：刘阳

学号：2213600

专业：信息安全法学双学位

指导教师：苏明

2024 年 10 月 13 日

目录

一、 比特币选用	1
二、 生成客户私钥	1
三、 ex2a——创建多签名交易	1
四、 ex2b——赎回多签名交易	4

一、 比特币选用

本次实验我继续使用第一次实验中将一比 bitcoin 均分产生的结果



二、 生成客户私钥

运行 keygen.py 获取三个客户的私钥和地址

Lab2 获取三个客户的私钥:

```

cust1_private_key= cQVsk59yrstUic9KuT9CbnyGDUeHJjuqwx1BmqH19huBnRtRPKT
address1 = mvEst63uib8cUpDijy6XvPAafGfERuLt8
cust2_private_key= cNZsz6BNQxTqRfvUFoxEnCsHvVckYKsx59uFomHJbZrBvWCMKQLC
address2 = n4WgGkATGJ2GpgtFREHZ76VtZKUQryUj5F
cust3_private_key= cU1JNcRBZf5Aa9hQDnYcS84zMHCmKAGnTnp1k8yQVPetCYSqT6Ri
address3 = myurbY8FNzM6T4emB4pZuXyafMKDjZTrc9
  
```

将客户私钥填入 ex2a.py 对应位置:

```

1 cust1_private_key = CBitcoinSecret(
2     'cQVsk59yrstUic9KuT9CbnyGDUeHJjuqwx1BmqH19huBnRtRPKT')
3 cust1_public_key = cust1_private_key.pub
4 cust2_private_key = CBitcoinSecret(
5     'cNZsz6BNQxTqRfvUFoxEnCsHvVckYKsx59uFomHJbZrBvWCMKQLC')
6 cust2_public_key = cust2_private_key.pub
7 cust3_private_key = CBitcoinSecret(
8     'cU1JNcRBZf5Aa9hQDnYcS84zMHCmKAGnTnp1k8yQVPetCYSqT6Ri')
9 cust3_public_key = cust3_private_key.pub
  
```

三、 ex2a——创建多签名交易

对于一个多签名交易锁定脚本, 有如下格式:

- M 操作码 (OP_CODE), 表示需要多少个签名来解锁这个输出。
- <PublicKey1>
- <PublicKey2>
- ...
- <PublicKeyN> 参与者的公钥, 按照一定**顺序**排列
- N 操作码 (OP_CODE), 表示总共有多少个公钥参与多重签名
- OP_CHECKMULTISIG 操作码, 用于验证签名。

在这里, 我规定除银行签名外, 还必须至少有两个签名, 因此 ex2a__scriptPubKey 设置为如下格式:

```

1 ex2a_txout_scriptPubKey = [
2     my_public_key,      # 银行的公钥
3     OP_CHECKSIGVERIFY,
4     OP_2,               # 需要两个签名
5     cust1_public_key,   # 第一个客户的公钥
6     cust2_public_key,   # 第二个客户的公钥
7     cust3_public_key,   # 第三个客户的公钥
8     OP_3,               # 总共三个公钥
9     OP_CHECKMULTISIG    # 多重签名检查
10 ]

```

设置参数如下:

```

1 # TODO: set these parameters correctly
2 amount_to_send = 0.00001
3 txid_to_spend = (
4     'e876a96965a3b816fb56f71b4f61c9fa038e4af35bfda3d9530f80f995b975cc ')
5 utxo_index = 2

```

运行 ex2a.py, 结果如下:

```

1 201 Created
2 {
3     "tx": {
4         "block_height": -1,
5         "block_index": -1,
6         "hash": "97928076fe31a5f41e289ba5023ca0bbb81a6596b68fa6e6c3e0a197cc4bb072",
7         "addresses": [
8             "mrbr3nvDsy1C9VgsJ9mGjW2GCphL6mPQwN",
9             "zJBoFj8AAufvAWgJSurn8kPXnkXyZMa7P"
10        ],
11        "total": 1000,
12        "fees": 2333,
13        "size": 306,
14        "vsize": 306,

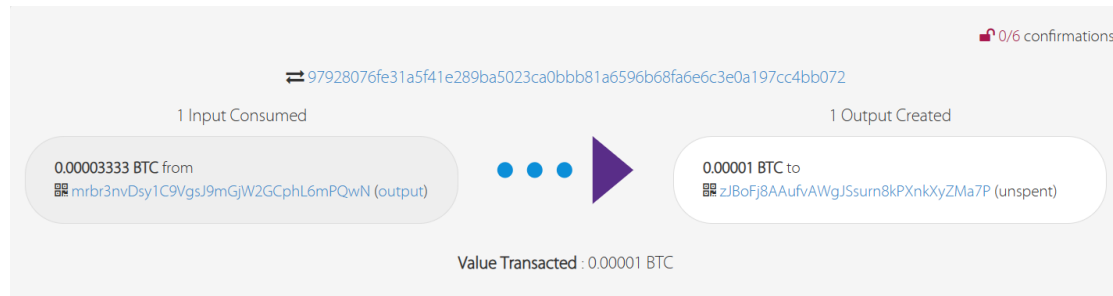
```

```

15     "preference": "low",
16     "relayed_by": "221.238.245.10",
17     "received": "2024-10-13T04:00:08.325099598Z",
18     "ver": 1,
19     "double_spend": false,
20     "vin_sz": 1,
21     "vout_sz": 1,
22     "confirmations": 0,
23     "inputs": [
24         {
25             "prev_hash": "
                e876a96965a3b816fb56f71b4f61c9fa038e4af35bfda3d9530f80f995b975cc
                ",
26             "output_index": 2,
27             "script": "473044022055
                e1934f9294828af53e14501001cfe474f0f3778715fe83a464c0
28             e54a633bbb0220241634dad23f98f88d10f63a2b0aa52c5e16443bb3f1c78ce73
29             b1cca664e99c2012102e75fc5d789372ece3634dff6d00b03d8b2d605940d908f
30             d37ce2e1ce32e8d5c6",
31             "output_value": 3333,
32             "sequence": 4294967295,
33             "addresses": [
34                 "mrbr3nvDsy1C9VgsJ9mGjW2GCphL6mPQwN"
35             ],
36             "script_type": "pay-to-pubkey-hash",
37             "age": 3071162
38         }
39     ],
40     "outputs": [
41         {
42             "value": 1000,
43             "script": "2102
                e75fc5d789372ece3634dff6d00b03d8b2d605940d908fd37ce2e1ce32e8d5c6a
                d522102dd1e17e6f6ddd3530a47433a3e950c91818d8d4688de7c20925476a318
                a09a1221026882c4080a307897f628e79d7ba57930c3c8c1558fbdcaaf3eab5ee
                b259006c221034b4b6ff5677acddeeea2bf87accd6195ac6733f7e37bd4f0e72e
44             3921e8d042af53ae",
45             "addresses": [
46                 "zJBofJ8AAufvAWgJSsurn8kPXnkXyZMa7P"
47             ],
48             "script_type": "pay-to-multi-pubkey-hash"
49         }
50     ]
51 }
52 }

```

在网站上可以看到生成了一个新的账户 **zJBoFj8AAufvAWgJSsurn8kPXnkXyZMa7P**, 供三个账户维护:



四、 ex2b——赎回多签名交易

根据上面对多重签名的要求, 我补充 `ex2b.py` 如下:

```

1 def multisig_scriptSig(txin, txout, txin_scriptPubKey):
2     bank_sig = create_OP_CHECKSIG_signature(txin, txout, txin_scriptPubKey,
3                                             my_private_key)
4     cust1_sig = create_OP_CHECKSIG_signature(txin, txout, txin_scriptPubKey,
5                                             cust1_private_key)
6     cust2_sig = create_OP_CHECKSIG_signature(txin, txout, txin_scriptPubKey,
7                                             cust2_private_key)
8     cust3_sig = create_OP_CHECKSIG_signature(txin, txout, txin_scriptPubKey,
9                                             cust3_private_key)
10    #####
11    # TODO: Complete this script to unlock the BTC that was locked in the
12    # multisig transaction created in Exercise 2a.
13    #scriptSig = [OP_0, cust3_sig, bank_sig]
14
15    return [OP_0, cust1_sig, cust2_sig, cust3_sig, bank_sig]
16    #####

```

设置参数如下:

```

1 # TODO: set these parameters correctly
2 amount_to_send = 0.00001
3 txid_to_spend = '97928076
4                 fe31a5f41e289ba5023ca0bbb81a6596b68fa6e6c3e0a197cc4bb072 '
5 utxo_index = 0

```

运行 `ex2b.py`, 结果如下:

```

1 201 Created
2 {
3     "tx": {
4         "block_height": -1,
5         "block_index": -1,

```

```

6      "hash": "361834405dcdaf20e16b933f3b3802716b1a65ddae7e50e712e06ed7ee1eb834
7      ",
8      "addresses": [
9          "zJBoFj8AAufvAWgJSsurn8kPXnkXyZMa7P",
10         "mv4rnyY3Su5gjcDNzbMLKBQkBicCtHUtFB"
11     ],
12     "total": 1000,
13     "fees": 0,
14     "size": 379,
15     "vsize": 379,
16     "preference": "low",
17     "relayed_by": "221.238.245.10",
18     "received": "2024-10-13T04:08:40.937655192Z",
19     "ver": 1,
20     "double_spend": false,
21     "vin_sz": 1,
22     "vout_sz": 1,
23     "confirmations": 0,
24     "inputs": [
25         {
26             "prev_hash": "97928076
27                 fe31a5f41e289ba5023ca0bbb81a6596b68fa6e6c3e0a197cc4bb072",
28             "output_index": 0,
29             "script": "0048304502210081
30                 ec9d09190ef6b5a9ad0d3f190fabf3e4d0bcf7f248540890799706f7fc2b42022
31                 0111
32                 dd9797360eb7c8d8ef0fe07cd3d202bce24b31f2205ab8d766e01c2e0986f
33                 01483045022100d19d0c48b58bb1440cea84006b393d7df52b8f7b82987877b25
34
35                 d5d6c4c6120770220267fb9c328f1105fe7cd8370220b0960326185f4bf54dd31
36                 74
37                 bc532afd2d8b150147304402204f88abe49995c67961b0691fbd86a99360c6a
38                 bb5c3059e4dfa7d2d37ce1871f5022039734bbd5c45d55a7e438d2f817dbd7fa9
39
30             "script_type": "pay-to-multi-pubkey-hash",
31             "age": 0
32         }
33     ],
34     "outputs": [
35         {

```

```
40     "value": 1000,  
41     "script": "76a9149f9a7abd600c0caa03983a77c8c3df8e062cb2fa88ac",  
42     "addresses": [  
43         "mv4rnyY3Su5gjcDNzbMLKBQkBicCtHUtFB"  
44     ],  
45     "script_type": "pay-to-pubkey-hash"  
46 }  
47 ]  
48 }  
49 }
```

查看网站得到如下结果:

